Gravitation

The big idea

The force that makes an apple fall also keeps the Moon in orbit. Isaac Newton realised this is one **universal** force: **gravitation**.

Universal law of gravitation

NEWTON'S LAW OF GRAVITATION

$$F = G m_1 m_2 / d^2$$

$G = 6.67 \times 10^{-11}$ N m^2 kg^{-2} (universal gravitational constant)

Every object attracts every other object with a force proportional to the product of their masses and inversely proportional to the square of the distance between them. Doubling the distance makes the force **one-fourth**. It explains tides, the Moon's orbit and planets around the Sun.

Free fall and g

Objects falling only under gravity are in **free fall** with acceleration **g ≈ 9.8 m/s^2** near Earth's surface, independent of their mass (ignoring air resistance; a feather and a coin fall together in a vacuum).

ACCELERATION DUE TO GRAVITY

$g = GM / R^2$, where M is Earth's mass and R its radius

g is slightly greater at the poles than at the equator, because Earth's radius is smaller at the poles. The equations of motion work with a = g.

Mass and weight

MASS VS WEIGHT

Mass is the amount of matter; it is the same everywhere (kg).
Weight is the force of gravity on an object: **W = mg** (N). It changes with g. The Moon's gravity is about **1/6** of Earth's, so you would weigh 1/6 as much there, with the same mass.

Thrust and pressure

Thrust: force acting perpendicular to a surface. **Pressure** = thrust / area (unit pascal, Pa = N/m^2). Sharp knives and nails have small areas (large pressure); buildings have wide foundations and school bags have broad straps (small pressure).

Fluids exert pressure in all directions.

Buoyancy and Archimedes' principle

When an object is immersed in a fluid, the fluid pushes it up with a **buoyant force (upthrust)**.

ARCHIMEDES' PRINCIPLE

A body wholly or partially immersed in a fluid experiences an upward force equal to the **weight of the fluid displaced**.
An object **floats** if its density is less than the fluid's, and **sinks** if it is greater.

Relative density = density of substance / density of water (no units). Uses: designing ships and submarines, lactometers (milk purity), hydrometers.

Quick recap

- $F = Gm_1m_2/d^2$; inverse-square law.
- $g \approx 9.8\ m/s^2$; $W = mg$; Moon's $g \approx g/6$.
- Pressure = thrust/area.
- Buoyant force = weight of fluid displaced.

Practice questions

1. If the distance between two objects is doubled, the gravitational force becomes
   (a) double (b) half (c) one-fourth (d) four times
2. The value of g on Earth's surface is about
   (a) 6.67 m/s² (b) 9.8 m/s² (c) 1.6 m/s² (d) 98 m/s²
3. An object weighs 60 N on Earth. Its weight on the Moon is about
   (a) 60 N (b) 10 N (c) 360 N (d) 6 N
4. The SI unit of pressure is
   (a) newton (b) pascal (c) joule (d) watt
5. An object floats in water if its density is
   (a) less than that of water (b) greater than that of water (c) equal to its weight (d) zero

Answers: 1. c 2. b 3. b 4. b 5. a